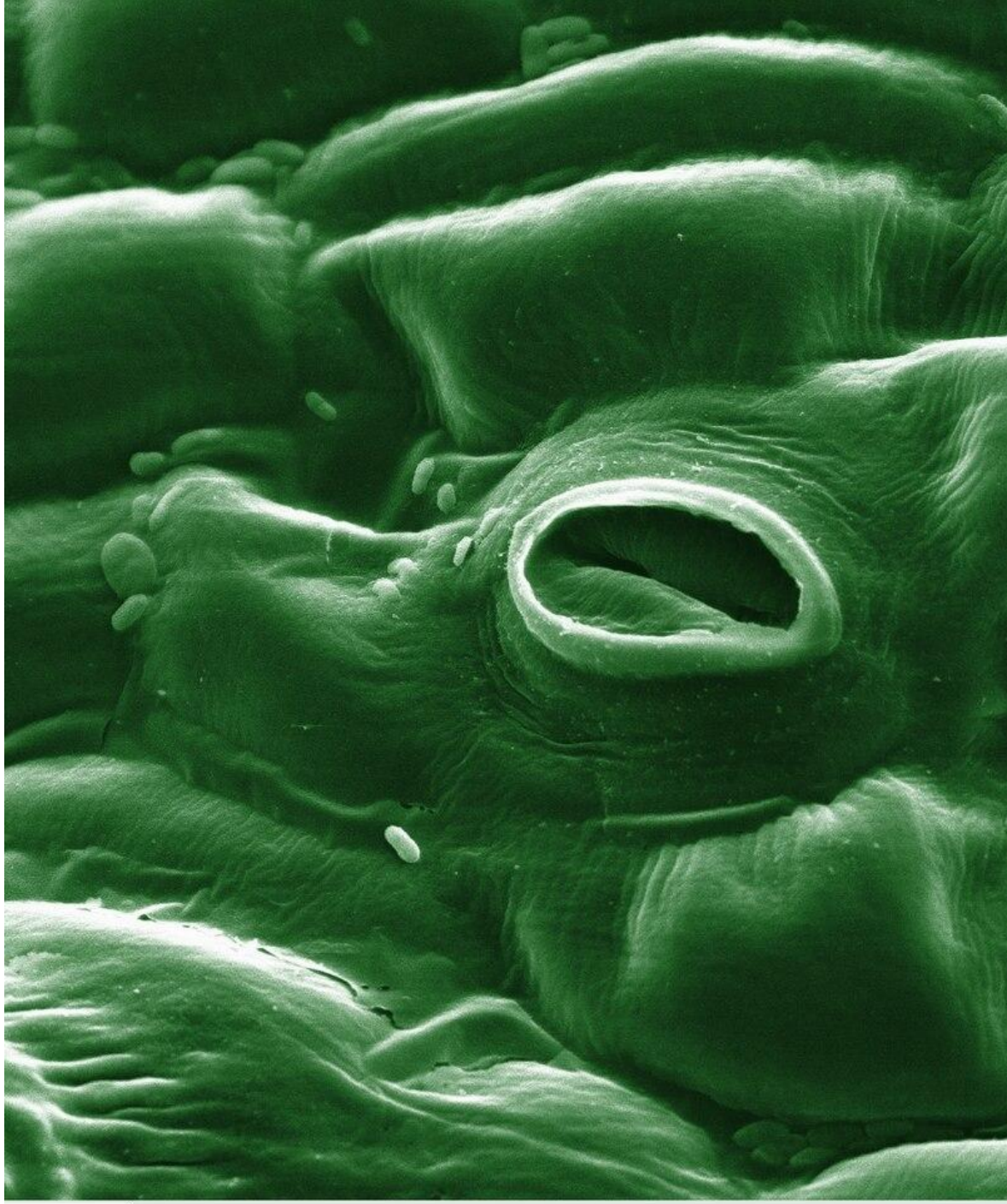
DP Biology – B3.1 Plant Systems

Gas exchange in plants

- What adaptations let multicellular organisms exchange gases?
- How do flowering plants and mammals differ, and resemble each other, in gas exchange?

Plant statements B3.1.7–10, SL and HL. Textbook pp. 283–291.
Image: a stoma on a tomato leaf, coloured electron micrograph. Photohound, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.1.7–8	Explain how leaf structures are adapted to gas exchange, and draw a plan diagram of a leaf
2	B3.1.9–10	Explain why gas exchange causes transpiration and what affects its rate, and determine stomatal density

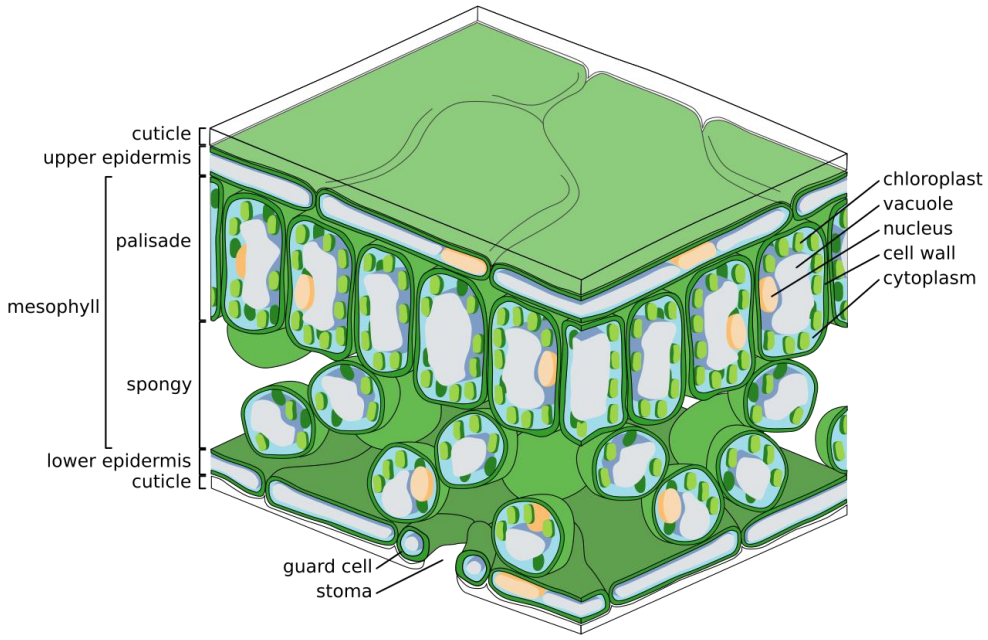
Overview of this chapter

Part	Statements	What it covers
The leaf	B3.1.7	How its tissues are adapted to gas exchange
Plan diagrams	B3.1.8	Where the tissues are in a dicot leaf
Transpiration	B3.1.9	Why it happens, and what affects its rate
Stomatal density	B3.1.10	Counting stomata from micrographs or leaf casts

Leaves need gas exchange too

B3.1.7

- Photosynthesis takes in **carbon dioxide** and releases **oxygen**
- Cell respiration does the reverse, all the time
- Gases diffuse through the air spaces inside the leaf
- Leaves are **flat and thin**, giving a large surface area-to-volume ratio and short diffusion distances



Tissues in a leaf. Zephyris, CC BY-SA 3.0.

Key idea

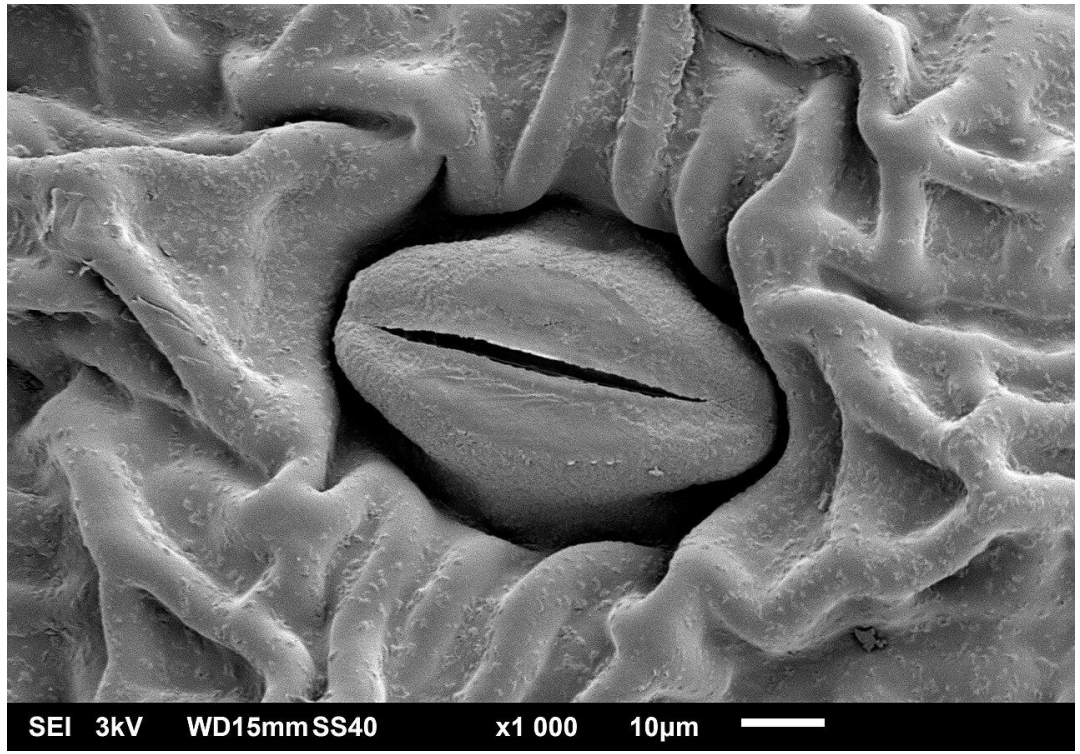
The same surface that lets carbon dioxide in also lets water vapour out. Every adaptation of a leaf balances these two.

How leaf structures help

B3.1.7

Structure	Adaptation
Waxy cuticle	Largely impermeable barrier, so gases and water vapour must pass through stomata, where the flow is controlled
Epidermis	Thin and transparent. Most pores are in the lower epidermis, where it is shaded and cooler, reducing water loss
Air spaces	Let gases circulate around loosely packed mesophyll cells and maintain gradients
Spongy mesophyll	Large, moist surface for gases to enter and leave cells
Stomatal guard cells	Open and close the pore to control gas exchange and water loss
Veins	Xylem brings water; phloem removes sugars, so gradients are maintained

Guard cells control the stoma



Stoma and guard cells on a leaf surface, electron micrograph. Judyta Dulnik, CC BY-SA 4.0.

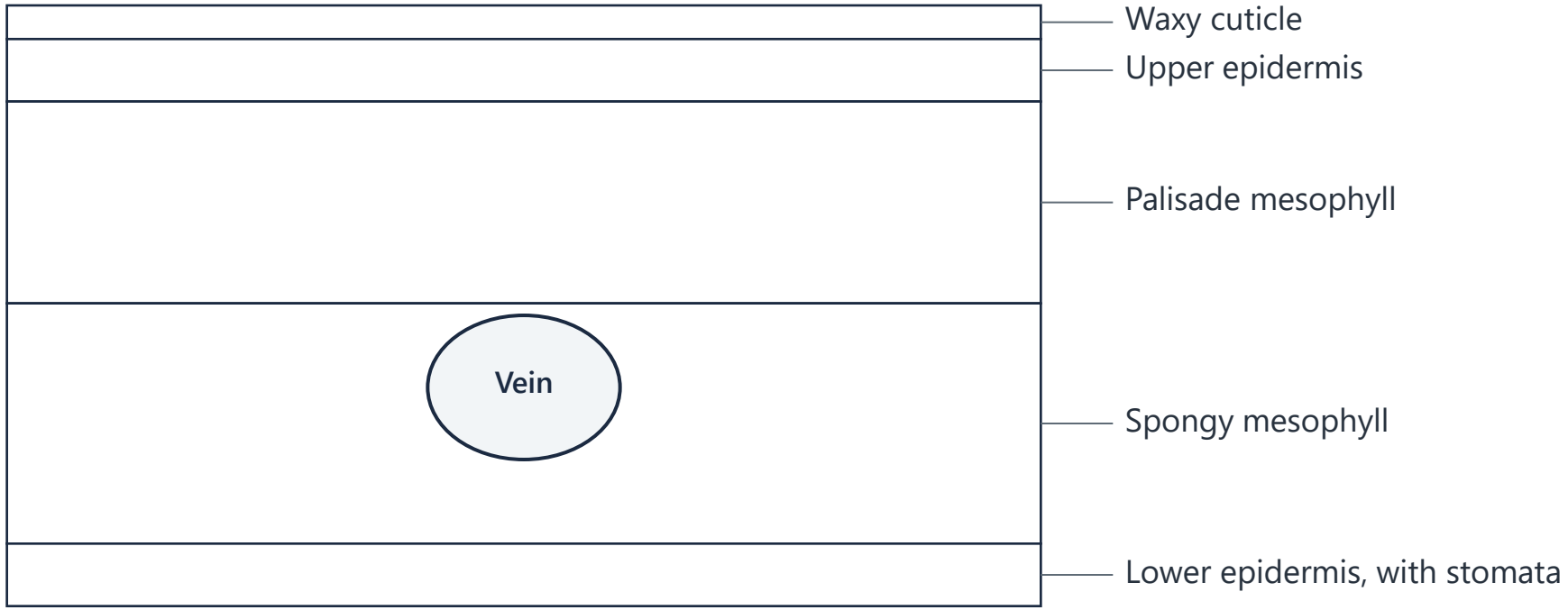
- Water enters the guard cells and they become **turgid**
- Thick inner walls make them curve apart, so the **pore opens**
- When they lose water they become **flaccid** and the **pore closes**
- Stomata usually open in daylight and close in the dark
- In very dry conditions, they close even in daylight

Common mistake

A stoma is the pore. Guard cells are the cells on either side of it, so do not use the two words as if they mean the same thing.

Plan diagram of a dicot leaf

B3.1.8



Schematic plan diagram of a transverse section.

Plan diagram

Show the tissues as regions. Do not draw individual cells or use shading.

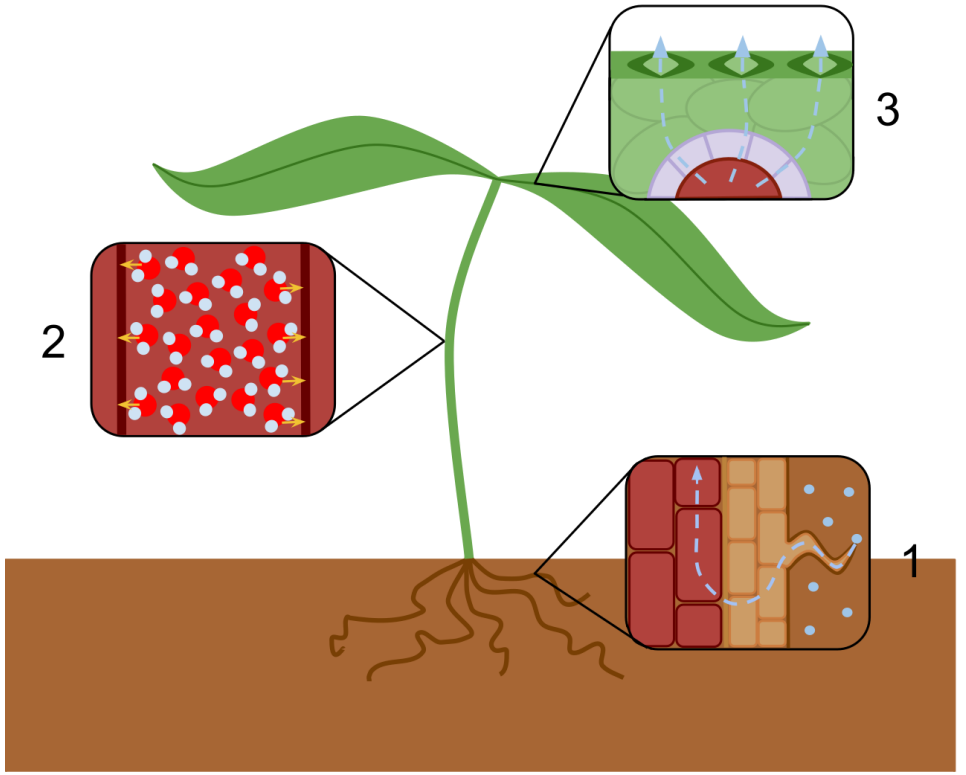
Activity

Draw a plan diagram of a transverse section from a micrograph. Use clear, continuous lines, label each tissue and keep the proportions of the original. Label the waxy cuticle, epidermis, palisade and spongy mesophyll, xylem and phloem.

Transpiration follows from gas exchange

B3.1.9

1. Stomata open so that carbon dioxide can enter
2. Water evaporates from moist cell walls into the air spaces
3. Water vapour diffuses out through the stomata, down its concentration gradient
4. This loss of water vapour from leaves is **transpiration**



Water moving through a plant. Laurel Jules, CC BY-SA 3.0.

Common mistake

Water vapour diffuses out of the leaf, not liquid water. Do not call this osmosis: the water is not crossing a cell membrane at this point.

What affects the rate of transpiration

B3.1.9

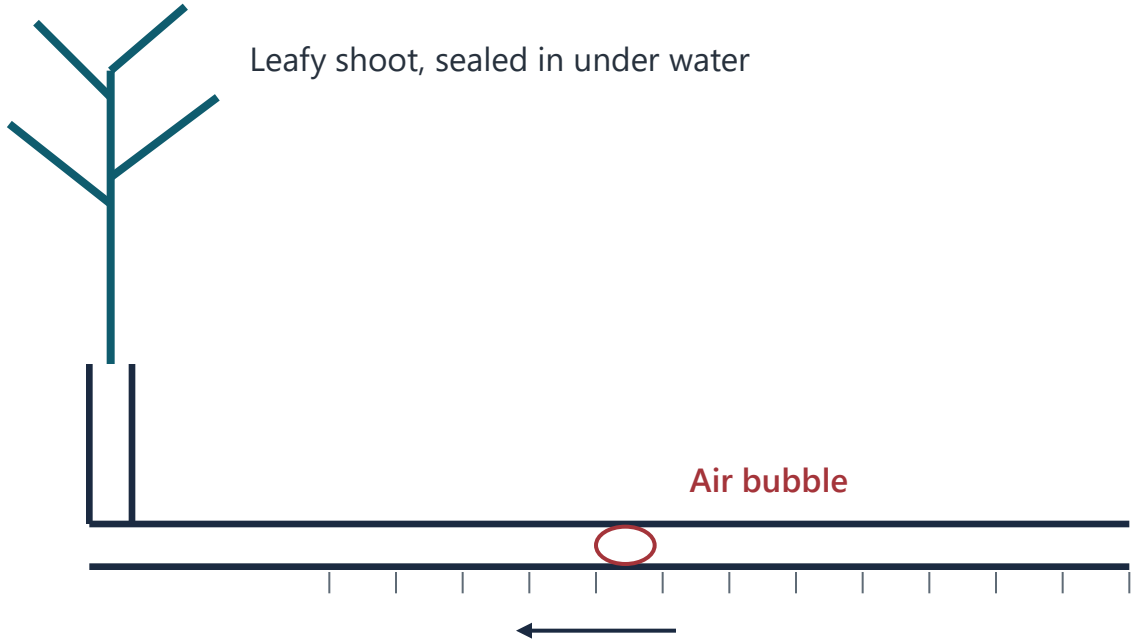
Factor	Effect of an increase	Why
Temperature	Rate rises	More evaporation, so a greater concentration difference of water vapour
Humidity	Rate falls	Air outside holds more water vapour, so the gradient is smaller
Wind	Rate rises	Sweeps away water vapour near the stomata, keeping the gradient steep
Light	Rate rises	Stomata open for photosynthesis, and the leaf warms

Exam note

When you explain any factor, refer to the concentration gradient of water vapour between the air spaces and the air outside.

Measuring transpiration: the potometer

B3.1.9



Capillary tube and scale. The bubble moves towards the shoot.

Schematic, not to scale.

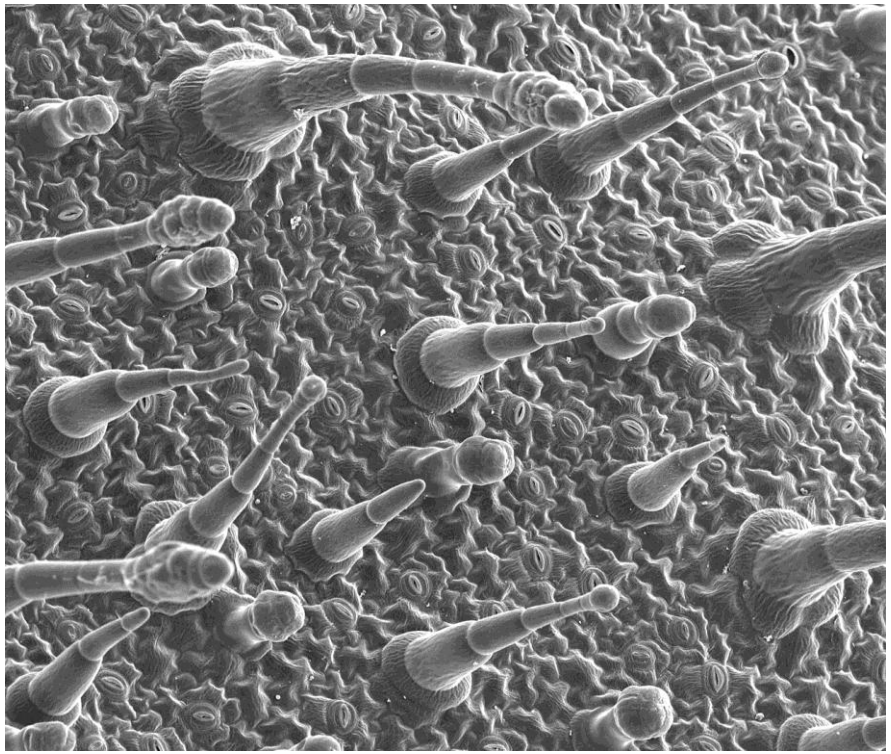
- Water lost from the leaves is replaced by water drawn up the stem
- The bubble moves along the tube as this happens
- **Rate** = distance moved / time
- Change one factor, such as wind from a fan or humidity from a plastic bag

Common mistake

A potometer measures water uptake, not transpiration directly. A small amount of the water is used in the plant, though for a small shoot the difference is minor.

Stomatal density

B3.1.10



Leaf epidermis, electron micrograph. Louisa Howard, Dartmouth, public domain.

- **Stomatal density**: the number of stomata per unit area of leaf
- Use a micrograph, or make a **leaf cast**
- Paint clear nail varnish on the underside; let it dry; lift it off with clear tape
- Count the stomata in a field of view, in several fields
- Area of field of view = 3.14 × r^2, and density = mean count / area

Why it varies

More stomata can take up more carbon dioxide, but they can also lose more water. Density varies between species and between environments.

Worked example: calculating density

B3.1.10

Field of view	Stomata counted
1	22
2	18
3	20

Invented values, for practice.

1. Mean count = (22 + 18 + 20) / 3 = 20
2. Field diameter = 0.4 mm, so r = 0.2 mm
3. Area = 3.14 × 0.2^2 = 0.126 mm^2
4. Density = 20 / 0.126 = 159 stomata per mm^2

Activity

Make leaf casts of the upper and lower surfaces of one leaf. Calculate the density for each surface. Explain the difference, and suggest how your results would differ for a plant from a dry habitat.

Comparing leaves and lungs

	Mammal lung	Leaf
Surface	Alveolar walls	Moist cell walls of mesophyll
Large area from	Many small alveoli	Air spaces around mesophyll cells
Gas moved in	Oxygen	Carbon dioxide (in the light)
Gradient kept by	Ventilation and blood flow	Photosynthesis, and air movement
Water loss limited by	Internal position of the lungs	Waxy cuticle and guard cells

Link
This answers the second guiding question. See B3.1.1–6 for the mammal side.

Check-in: gas exchange in leaves

B3.1.6–8

1. Give three features of a leaf that help gas exchange.
2. Explain how guard cells open a stoma.
3. Explain why transpiration is a side effect of gas exchange.
4. State three factors that increase the rate of transpiration and explain one.
5. Outline how a potometer is used.
6. Explain why stomatal density is usually higher on the lower surface of a leaf.

How to use this

Answer without your notes, then compare with a partner.

Working with data: a potometer experiment

B3.1.8

Distance moved by the bubble in 5 minutes. These data are invented for practice; the pattern matches real results.

Condition	Distance / mm	Rate / mm min^{-1}
Still air, room light	20	?
Fan on	45	?
Plastic bag over shoot	6	?
Lamp on	34	?

1. Calculate each rate.
2. Explain the result with the fan.
3. Explain the result with the bag.
4. State one assumption made when a potometer is used.

Common mistakes to avoid

Not this	But this
Plants do not respire, so they only take in carbon dioxide.	Plants respire all the time; in the light, photosynthesis is usually faster.
Transpiration is evaporation from the leaf surface.	Most water evaporates inside the leaf and diffuses out through stomata.
Guard cells open by losing water.	They open when they gain water and become turgid.
A potometer measures transpiration exactly.	It measures water uptake.

Check that you can...

1. **Explain** how leaf structure suits gas exchange
2. **Draw** and **label** a plan diagram of a dicot leaf
3. **Explain** how stomata open and close
4. **Explain** how environmental factors affect transpiration
5. **Calculate** stomatal density and rates from a potometer
6. **Compare** gas exchange in leaves and lungs

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why gas exchange in a leaf leads to water loss.

2. Predict the effect of higher humidity on the rate of transpiration, and explain why.

3. A student counts a mean of 15 stomata in a field of view with an area of 0.15 mm^2. Calculate the stomatal density.

Image credits

Tomato leaf stomate 1-color.jpg – Photohound – Public domain
Leaf Tissue Structure.svg – Zephyris – CC BY-SA 3.0
Stoma.jpg – Judyta Dulnik – CC BY-SA 4.0
Transpiration Overview.svg – Laurel Jules – CC BY-SA 3.0
Leaf epidermis.jpg – Louisa Howard (Dartmouth electron microscope facility) – Public domain

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